

Unit 8: Basic Mineral Development

Reading: pages 115–124 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. What determines which element an atom is?
 - a) The number of protons in the nucleus
 - b) The number of electrons in orbit
 - c) The total mass of the atom
 - d) The number of neutrons in the nucleus
2. An atom that loses an electron becomes
 - a) a neutron
 - b) a compound
 - c) an anion with a negative charge
 - d) a cation with a positive charge
3. Which group of minerals makes up most of Earth's crust?
 - a) Carbonates
 - b) Native elements
 - c) Silicates
 - d) Oxides
4. Streak is considered a more reliable identification property than color because
 - a) streak does not vary
 - b) every mineral produces a streak
 - c) streak also measures hardness
 - d) streak is easier to observe than color
5. On the Mohs scale of mineral hardness, which mineral is the softest?
 - a) Gypsum
 - b) Talc
 - c) Calcite
 - d) Diamond

6. Mica breaks along very closely spaced flat planes that yield thin sheets. This property is called
- a) fracture
 - b) cleavage
 - c) streak
 - d) luster

Part B · True or False

2 questions × 1 point = 2 points

7. Color is usually the most reliable physical property for identifying a mineral.
TRUE / FALSE
8. Fracture is not always the same within a single mineral, because fracture is not determined by the mineral's structure.
TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. The International Mineralogical Association sets requirements a substance must meet to be considered a distinct mineral. Name three of them.

10. Explain the difference between *cleavage* and *fracture*, and give one example of a mineral that shows each.

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